

AI-based Personal Posture and Ergonomics Coach



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ACKNOWLEDGEMENT

All praise and gratitude are due to Allah Almighty, whose countless blessings, guidance, and strength have enabled me to complete this work successfully. Without His mercy, nothing would have been possible. I would like to extend my sincere appreciation to Dr. Syed Muhammad Naqi for his invaluable support, encouragement, and insightful guidance throughout this journey. His expertise and mentorship have been instrumental in shaping my understanding and approach. I am deeply grateful to my family for their unwavering love, patience, and constant motivation. Their support has been the foundation of my perseverance and success. Special thanks to our departmental Head, Dr. Ayaz Hussain, for providing a conducive learning environment and valuable insights that helped me throughout this process.

I am immensely thankful to Ma'am Onaiza for teaching us Software Construction, providing us with essential knowledge that proved crucial in this journey. I also wish to express my gratitude to my esteemed teachers—Sir Rabi, Sir Mudassir, Sir Akmal Khattak, Sir Muazzam Khattak, Ma'am Memoona, Sir Ali Naqi, and Sir Ghazanfar—for their dedication and commitment to teaching us programming, enabling us to achieve this milestone.

I acknowledge everyone who has contributed, directly or indirectly, to this achievement. Your support has been truly invaluable.

ABSTRACT

In today's world, prolonged computer usage has increased significantly among office workers, remote employees, and students who are required to sit in front of screens for extended hours on a daily basis. Continuous sitting in improper posture such as slouching, forward head position, or leaning leads to physical discomfort including neck pain, shoulder pain, upper and lower back discomfort, and muscle fatigue which ultimately reduces work efficiency and productivity. Most individuals remain unaware of their bad sitting habits until physical symptoms begin to appear. Existing posture correction solutions either require specialized hardware or wearable devices which are expensive and inconvenient for everyday use. Despite the growing need for accessible posture monitoring, there is a lack of simple, non-intrusive, and real-time solutions that work without any additional equipment.

AI-Based Personal Posture and Ergonomics Coach is a web application that helps users monitor and correct their sitting posture in real time using only a laptop's built-in webcam. It continuously captures the user's posture, detects key skeleton points using AI, and classifies posture as good or bad. When a user maintains bad posture beyond a predefined threshold of 60 seconds, the system generates a real-time alert and provides ergonomic recommendations to correct it immediately. The system also maintains a detailed posture log and generates weekly reports showing total monitoring duration, bad posture percentage, most frequent bad posture type, and overall session statistics so users can track their improvement over time. A posture history view allows users to review their past seven days of data. The solution requires no special hardware or wearable device, making it suitable and accessible for everyday use by students, office workers, and remote employees.

Looking ahead, the system aims to expand its capabilities by incorporating predictive posture analysis, personalized ergonomic coaching plans, and multi-platform support. With continuous development, the AI-Based Personal Posture and Ergonomics Coach envision becoming an essential productivity tool that promotes healthier work habits and reduces the long-term physical impact of prolonged computer usage.

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CHAPTER NUMBER 1

SOFTWARE PROJECT MANAGEMENT PLAN

Chapter 1: Software Project Management Plan

1.1 Introduction:

This first chapter introduces the description of “AI-based Personal Posture and Ergonomic Coach” what is the problem statement, and how it will be designed and developed to solve the particular problem. This chapter will also explain the scope, requirements and objectives of the project.

1.2 Problem Statement:

Today the use of computers has increased a lot in the lives of office workers, remote employees and students. As a result, individuals are required to sit in front of it for a long time. Continuous sitting in improper posture such as slouching, forward head or leaning can lead to physical discomfort and reduced work efficiency. Most Individuals are unaware of their bad posture sitting until they start feeling physical discomfort such as neck pain, shoulder pain, upper back discomfort, lower back discomfort, muscle fatigue.

Existing bad posture solutions may require either specialized hardware or wearable device. these approaches can be expensive or inconvenient for everyday use.

Therefore, there is a need for simple, non-intrusive and real-time posture monitoring system. So, the proposed “AI-Based Personal Posture & Ergonomics Coach” system will monitor user’s posture in real time and give recommendations for ergonomic improvement.

1.3 Project Description:

The proposed “AI-Based Personal Posture & Ergonomics Coach” is a web application, which observes user’s posture continuously through laptop’s webcam, detect key skeleton points and classify posture as good or bad. When user maintains bad posture for a duration which exceeds predefined threshold time (60 seconds), system generates an alert in real time and suggests ergonomic improvement accordingly.

The system also maintains a log, and based on these statistics, weekly reports can be generated which helps user understand his bad posture over time and make improvements to reduce physical discomfort and increase productivity. This solution is simple, non-intrusive and does not require any special hardware or wearable device. Which makes it suitable for everyday use.

1.4 Major Functionalities:

1. Signup user
2. Login User

3. Track posture
4. generate report(weekly)
5. View posture History(weekly)
6. Logout

1.5 Objectives:

Here are the main Objectives:

1. To help users to maintain ergonomic posture without requiring special hardware or wearable device.
2. To increase work productivity by reducing physical discomfort caused by prolonged poor sitting posture.
3. To promote healthy sitting posture while working on computer.
4. To address the posture-related issues that are becoming increasingly common among students, office workers, and remote employees.

1.6 Tools and techniques:

It is web-based application and following tools has been used in development

1. React
2. Nodejs
3. Express
4. Mongodb
5. MS word

1.7 Users:

1. Office workers: who works on computer for hours
2. remote employees or freelancers: who work from home using laptops
3. students: who study long hours on laptop

1.8 Software Process Model:

Waterfall Model will be used in the project because:

1. It has clear Requirements

1.9 Project Management Plan:

This section will elaborate on how the task will be divided among different people and resources.
What will be the deliverables, risks and constraints?

1.10 Requirements Analysis Phase:

1.10.1 Requirement Identification:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
The initial step of this project is the identification of requirements which are functional and nonfunctional.	Requirements are collected and reviewed	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word Hardware: Laptop	None	None

Table 1.1 Requirements Identification Task

1.10.2 Define Use Cases:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
After clearing all requirements software architecture will start with Use case diagrams	Use Cases are written and then should be submitted as a Use-Case Diagram in Word format	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word, draw.io Hardware: Laptop	Requirements must be clear.	Unclear requirements

Table 1.2 Defining Use Cases Task

1.10.3 Develop Domain Model:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
After we know about the use cases of a project, we will start designing Domain Models for the system	Domain model will be reviewed	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word, draw.io Hardware: Laptop	We must know the use cases before we move to the Domain Model	ill-defined use cases

Table 1.3 Domain Model Development Task

1.10.4 ERD Designing:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we make ERD Designs for the database	ERD Design will be reviewed	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word, draw.io Hardware: Laptop	We must know the design of the domain model so we can see dataflow and what data we will need in this step	Any error in a Domain model

Table 1.4 ERD Designing Task

1.10.5 SRS Completion:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
This task includes making software requirements specification document	SRS document in Word or pdf form Requirement analysis completed	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word Hardware: Laptop	None	None

Table 1.5 SRS Completion Task

1.11 Designing Phase:

1.11.1 Develop Design:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
Development of architectural design	Architectural Diagram in PDF or Word format	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word, draw.io Hardware: Laptop	Requirement analysis must be completed	Ill-defined requirements

Table 1.6 Design Development Task

1.11.2 Develop Interfaces:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
This task includes how each screen looks like and what feature will come on what screen so we will make interfaces.	Canva Design, or Pdf format design	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS word, Canva Hardware: Laptop	Architectural design must be completed	Ill-defined requirements, Use cases

Table 1.7 Interface Development Task

1.11.3 Class Diagram Designing:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we create a Class Diagram to show the flow of data and how they relate to each other.	Class Diagram in PDF or Word format	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word, draw.io Hardware: Laptop	Architectural Design must be completed	ill-defined architectural design, and domain model.

Table 1.8 Class Diagram Task

1.11.4 Sequence Diagram Designing:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we make a Sequence Diagram to show the flow of data	Sequence Diagram in PDF or Word format	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word, draw.io Hardware: Laptop	Architectural Design must be completed	ill-defined architectural design, and domain model.

Table 1.9 Sequence Diagram Task

1.11.5 Design Phase Verification:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we make sure that all the steps completed before the previous phase are correct. We do previous phase verification.	Design phase is completed it's document will be reviewed.	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word, draw.io Hardware: Laptop	The whole Design phase must be completed	None

Table 1.10 Design Phase Verification Task

1.11.6 Software Test Documentation:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we define the test cases	Test cases are reviewed. Design phase is complete	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word Hardware: Laptop	Design phase should be complete	None

Table 1.11 Software Test Documentation Task

1.12 Implementation Phase:

1.12.1 System Implementation:

Task Description	Deliverables	Resources Needed	Dependencies & Constraints	Risks
In this step, we develop the web application	Complete web application source files including images, videos or anything embedded in it submitted in zip.	People: Hubab Masood Chaudhary, Dr Syed Muhammad Naqi Software: MS Word, VS Code, Photoshop, Illustrator, MySQL, MongoDB Hardware: Laptop	The whole Design phase must be completed	None

Table 1.12 System Implementation

1.13 Gantt Chart:

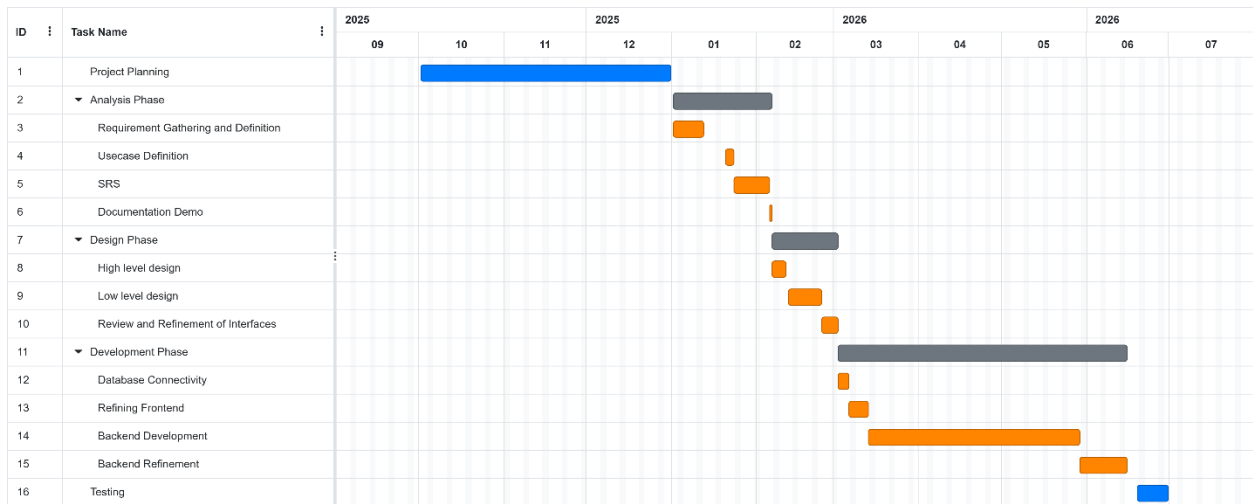


Figure 1 Gantt Chart

CHAPTER NUMBER 2

SOFTWARE REQUIREMENTS SPECIFICATIONS

2.1 Introduction:

This chapter first describes the attributes of the software system. It highlights the user characteristics and major constraints of the tool. It also elaborates on the use cases and system sequence diagrams. Finally, this chapter explains the domain model and database design of the project.

2.2 Functional Requirements:

1. Signup: When user opens my desktop app for the first time, he will have to signup by providing the asked information
2. Login: Each time already signed up user open my app, he will have to login using credentials.
3. Track posture: User will click on track posture and app will start monitoring his posture in real time.
4. Generate Posture report: Allows the user to see his posture detail weekly
5. View posture history: Allow the user to view read only posture history for last 7 days.

Functions (not use case)

6. Generate alert: when duration of bad posture exceeds threshold time(1 min), the system will generate real time alert to notify the user. If user does not correct posture, system will repeat an alert after every 2 mins until posture is corrected
7. Generate ergonomic recommendation: The system will provide recommendations for ergonomic improvement to correct the posture.
8. display posture graph: system will generate real time posture graph (simple line graph) when user clicks on “track posture”.
9. Backup of the system: After 24 hours, system will automatically take backup of the data.

2.3 Non-Functional Requirement:

1. Accuracy in Posture Detection
2. Real time Data Storage
3. Real time posture detection and alert generation (also ergonomic suggestion)

2.4 Constraints:

This system is only for laptops.

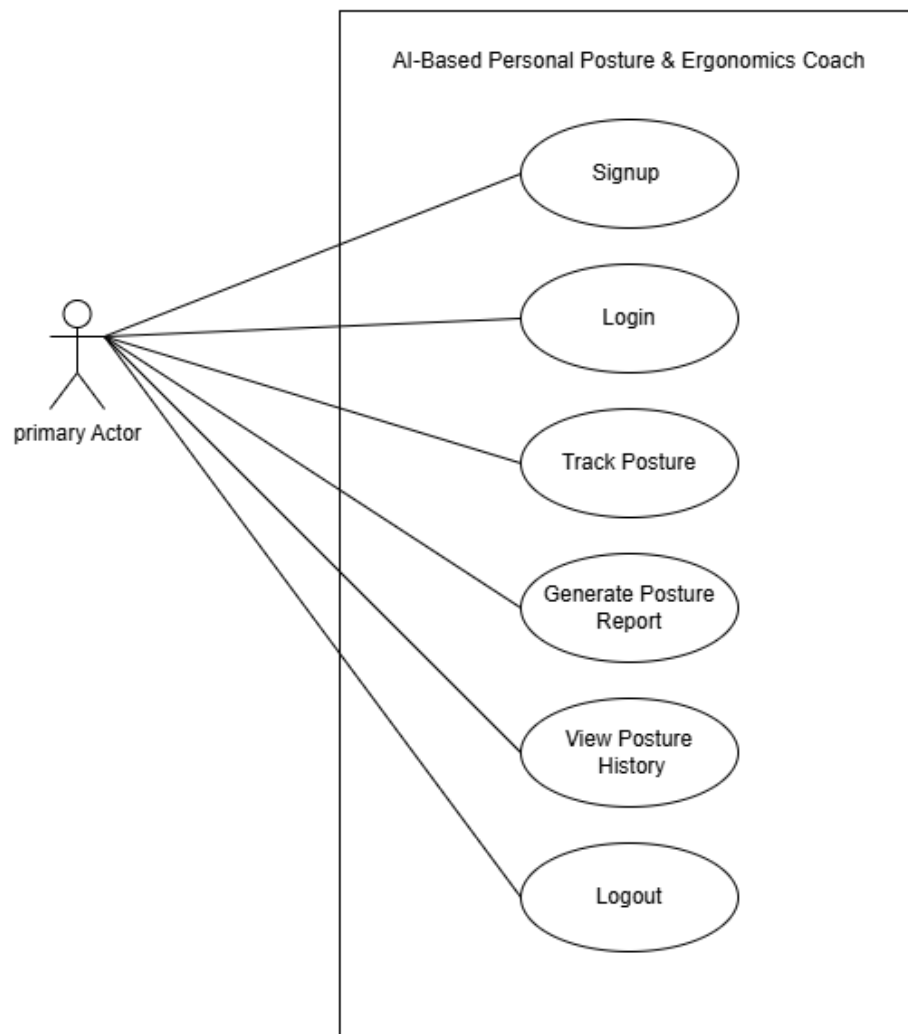
2.5 Major Input and Output:

Major Inputs:

1. Signup and login credentials (username, password, email)
2. Real time posture frames

Major Outputs:

1. Alert for bad posture
2. ergonomic suggestion
3. real time posture line graph
4. report and history output(total monitoring duration, total bad posture duration, bad posture percentage, good posture duration, most frequent bad posture, date, posture type)

2.6 Use Case Diagram:

2.7 Use Case Text:

2.7.1 Use Case text for “Signup”

ID: UD1

Name: Signup

Primary Actors: User (office workers, remote employees, students)

Pre-condition: None

Post-condition:

1. User’s account is created successfully
2. Main dashboard of app is displayed

Main Success scenario:

1. User requests for sign up
2. System shows signup page and asks for information
3. User enters the asked information
4. System creates user’s account and displays main dashboard of app

Alternative flow:

- 3a. User does not provide complete information
System asks user to complete required information
User provides missing complete information

2.7.2 Use Case text for “Login”

ID: UD2

Name: Login

Primary Actors: User (office workers, remote employees, students)

Pre-condition: User has already an account

Post-Condition:

1. User has been logged in
2. Main dashboard has been displayed

Main Success Scenario:

1. user requests for login
2. System shows login page and asks for credentials
3. User enters asked credentials
4. System logs in the user and displays main app dashboard.

Alternative flow:

- 3a. user enters false credentials

system displays an error

User re-enters correct credentials

3b. User does not provide complete information

System asks to enter required fields

User provides missing information

2.7.3 Use case text for “Track Posture”

ID: UC3

Name: Track Posture

Primary Actors: User (office workers, remote employees, students)

pre-conditions:

1. User is logged in.
2. Laptop’s webcam is on and working correctly.

post-conditions: User’s good posture is detected, bad posture time has been reset and monitoring again has been started

Main Success Scenario (Basic Flow):

1. User requests for “track posture”
2. The system starts capturing the user’s postures in real time
3. User sits in bad posture longer than the defined threshold time
4. The system detects bad posture
5. The system generates an alert in real time
6. The system provides ergonomic suggestion to correct the posture in real time
7. User corrects his posture

Alternate Flow:

- 6a.** User does not correct his position

The system repeats an alert after every 2 mins until posture is corrected

Special Requirement:

Alert should be non-disturbing and has no sound. If user is using another app, it should display on top of app without disturbing user activity

2.7.4 Use case text for “Generate Posture Report”:

ID: UD4

Name: Generate Posture report

Primary Actors: User (office workers, remote employees, students)

Pre-condition: User is logged in

post-condition: One week posture report is displayed

Main Success Scenario (Basic Flow):

1. User requests to generate a report
2. User selects a week (by default current week is selected)
3. System shows the report for that week

Alternative Flow:

- 1a. if user using the application for less than one week

System shows “Not sufficient data to generate report, kindly wait until one week is completed”

- 2a. If user wants to request for a week beyond current week or before the first available week.

The System does not allow to request for it

2.7.5 Use case text for “View Posture History”:

ID: UD5

Name: View Posture history

Primary Actors: user (office workers, remote employees, students)

Pre-condition: User is logged in

post-condition:

1. Current day history has been displayed
- or
2. one week history is displayed

Main Success Scenario (Basic Flow):

1. User requests to view history
2. System asks for current day or weekly history
3.
 - 3.1 User requests for current day history
 - 3.2 System shows current day history so far
4.
 - 4.1 User requests for weekly history
 - 4.2 User selects week (by default current week is selected)
 - 4.3 System shows selected week history

Alternative Flow:

4.2a If user wants to request for a week beyond current week or before the first available week
The System does not allow to request for it

2.7.6 Use case text for “Logout”:

ID: UD6

Name: Logout

Primary Actors: user (office workers, remote employees, students)

Pre-condition: User is logged in

post-condition:

1. User has been logged out.
2. Login/signup page appears

Main Success Scenario (Basic Flow):

1. User requests for logout
2. System asks for confirmation
3. User navigates to “Yes”.
4. System logout the user.

Alternative Flow:

3a. User selects “NO”

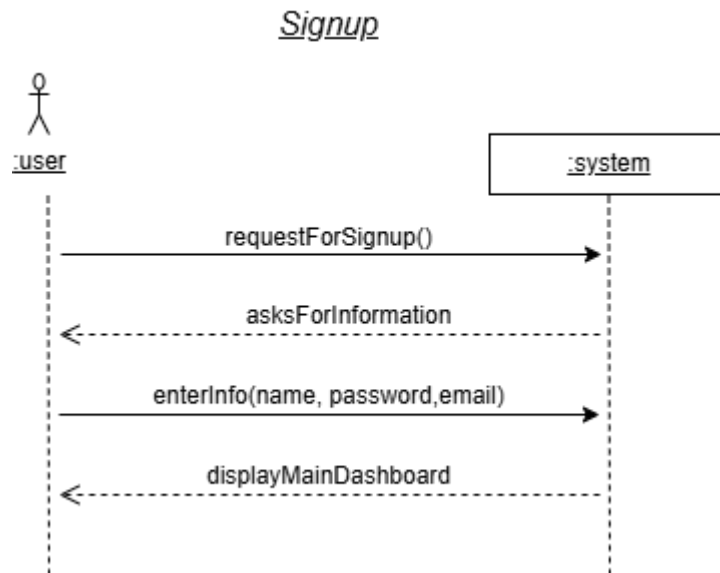
System cancels the logout and remain on same page

3b. User dismisses the confirmation box

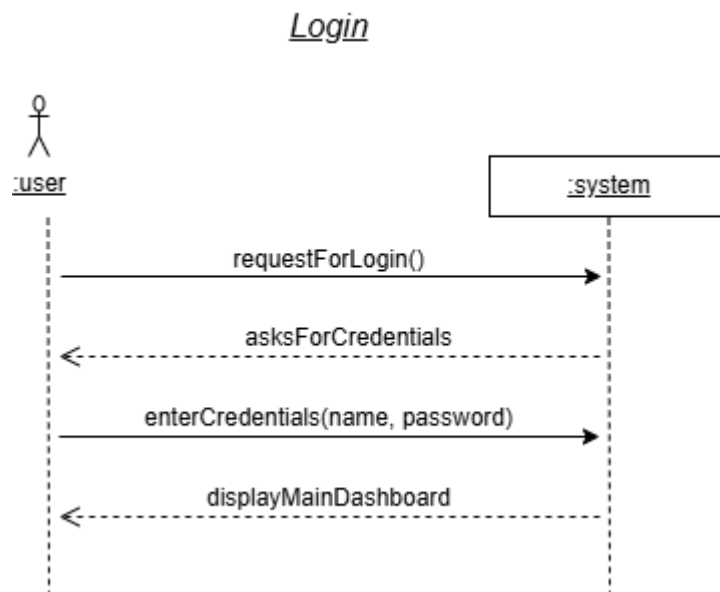
System cancels the logout and remains on dashboard

2.8 System Sequence Diagrams:

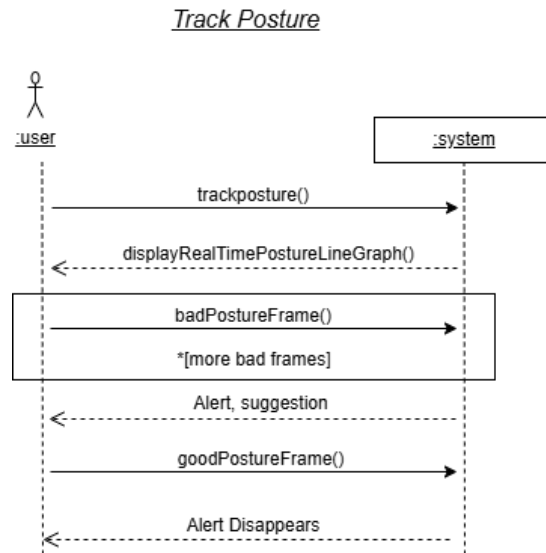
2.8.1 SignUp



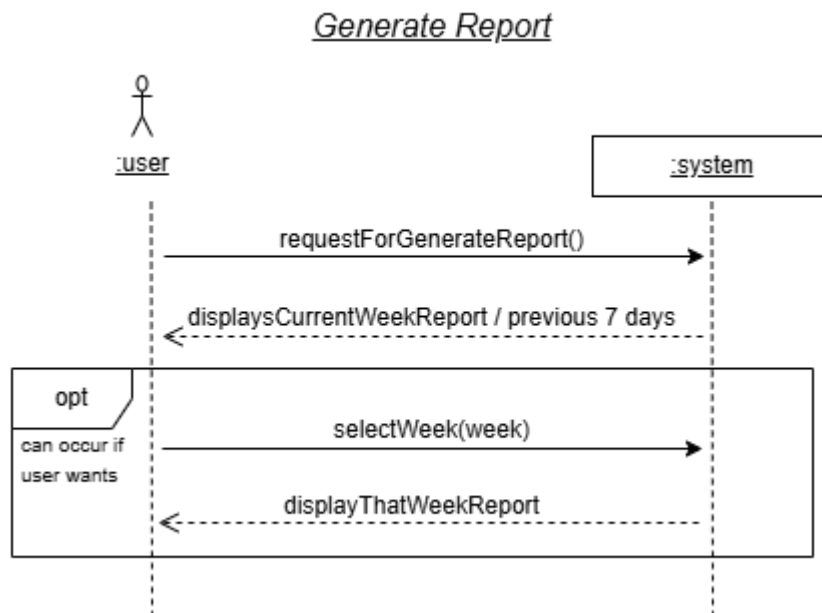
2.8.2 Login



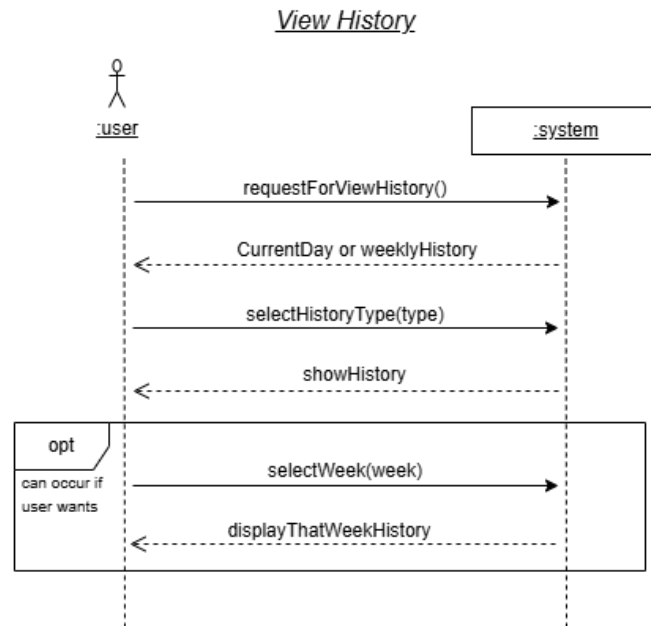
2.8.3 Track Posture



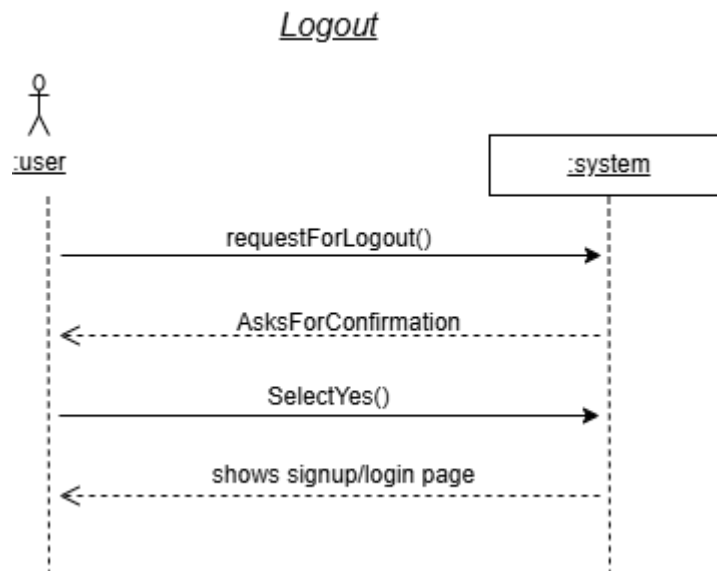
2.8.4 Generate Posture Report



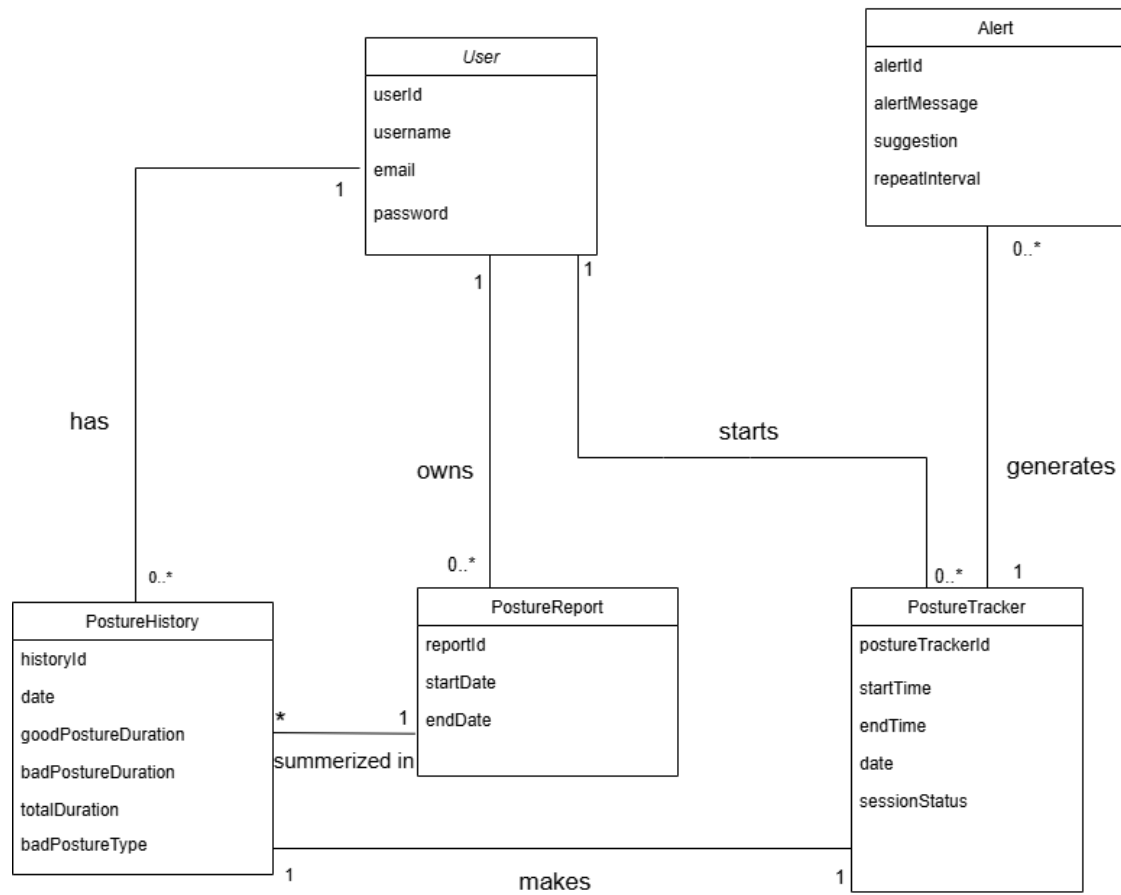
2.8.5 View Posture History



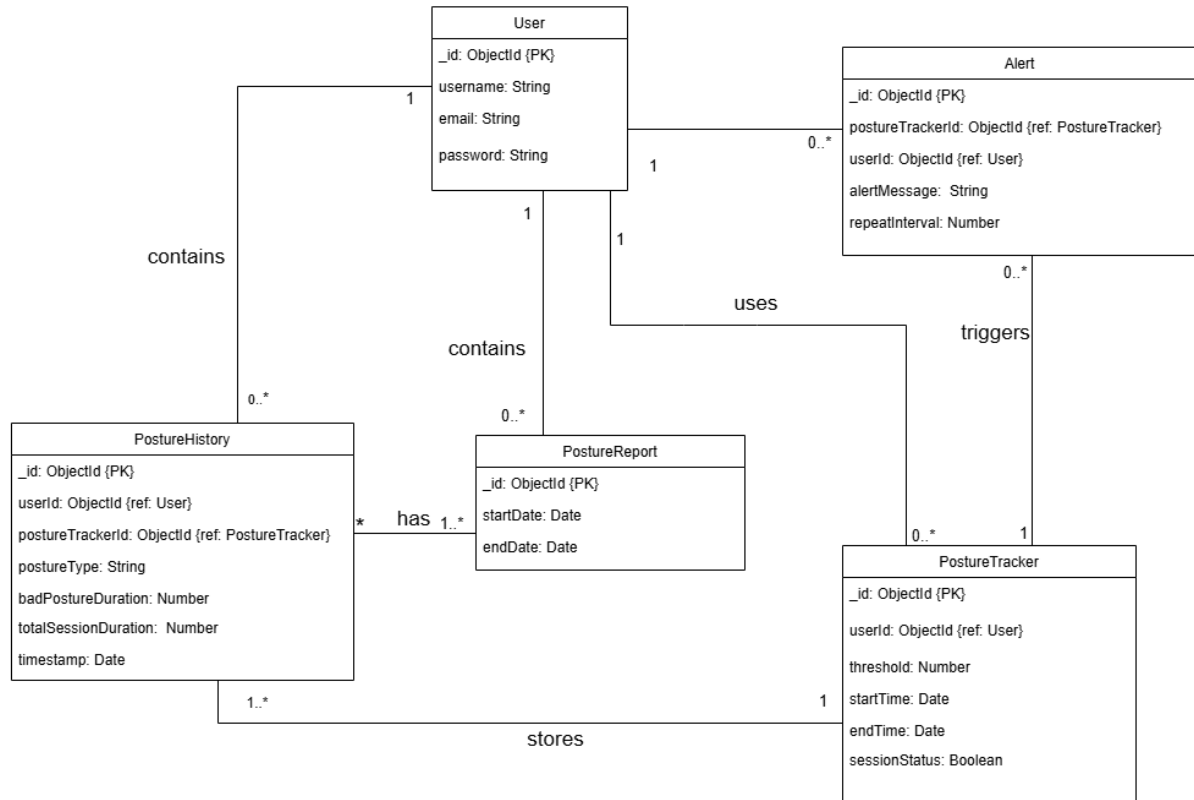
2.8.6 Logout



2.9 Domain Model:



2.10 ERD:



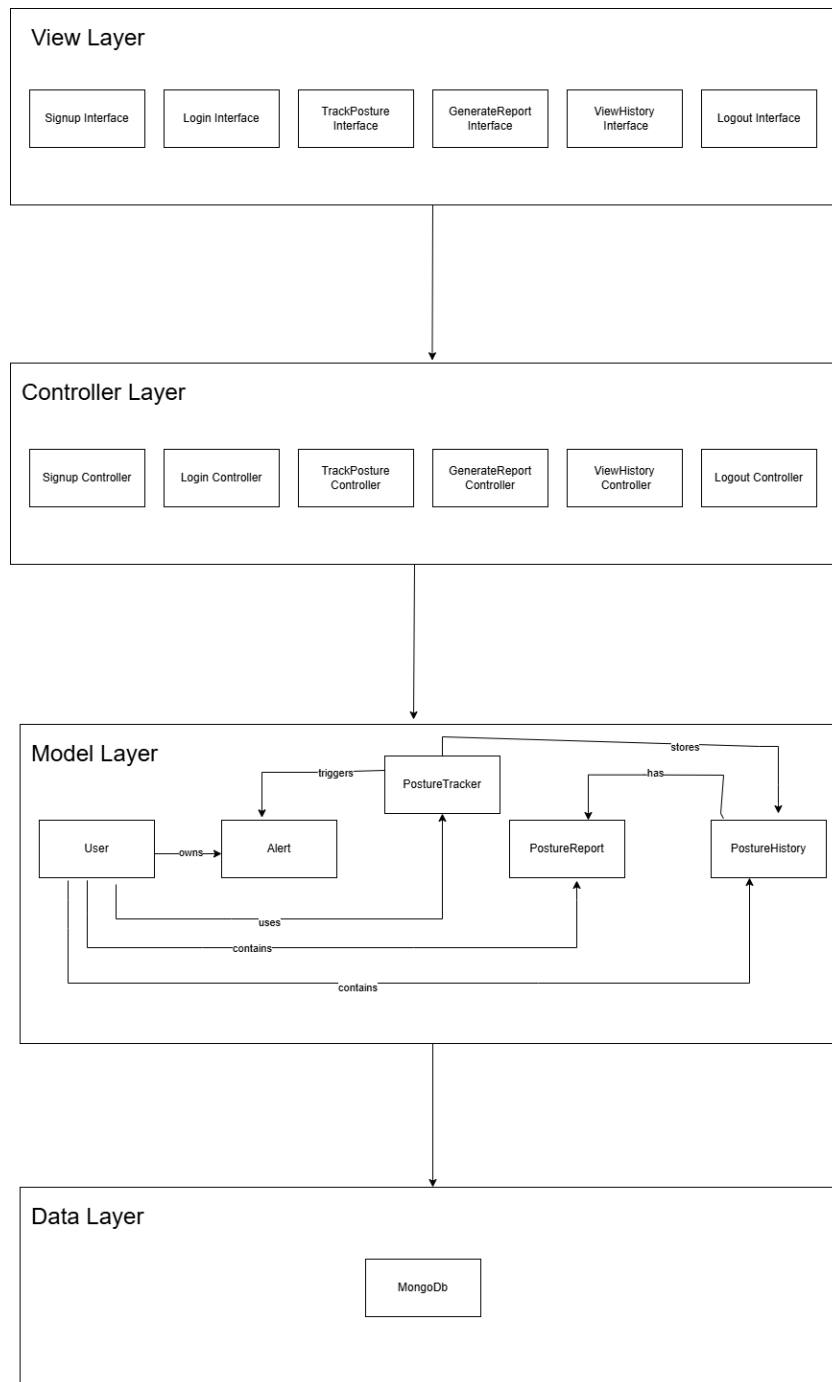
CHAPTER NUMBER 3

SOFTWARE DESIGN DESCRIPTION

3.1 Introduction

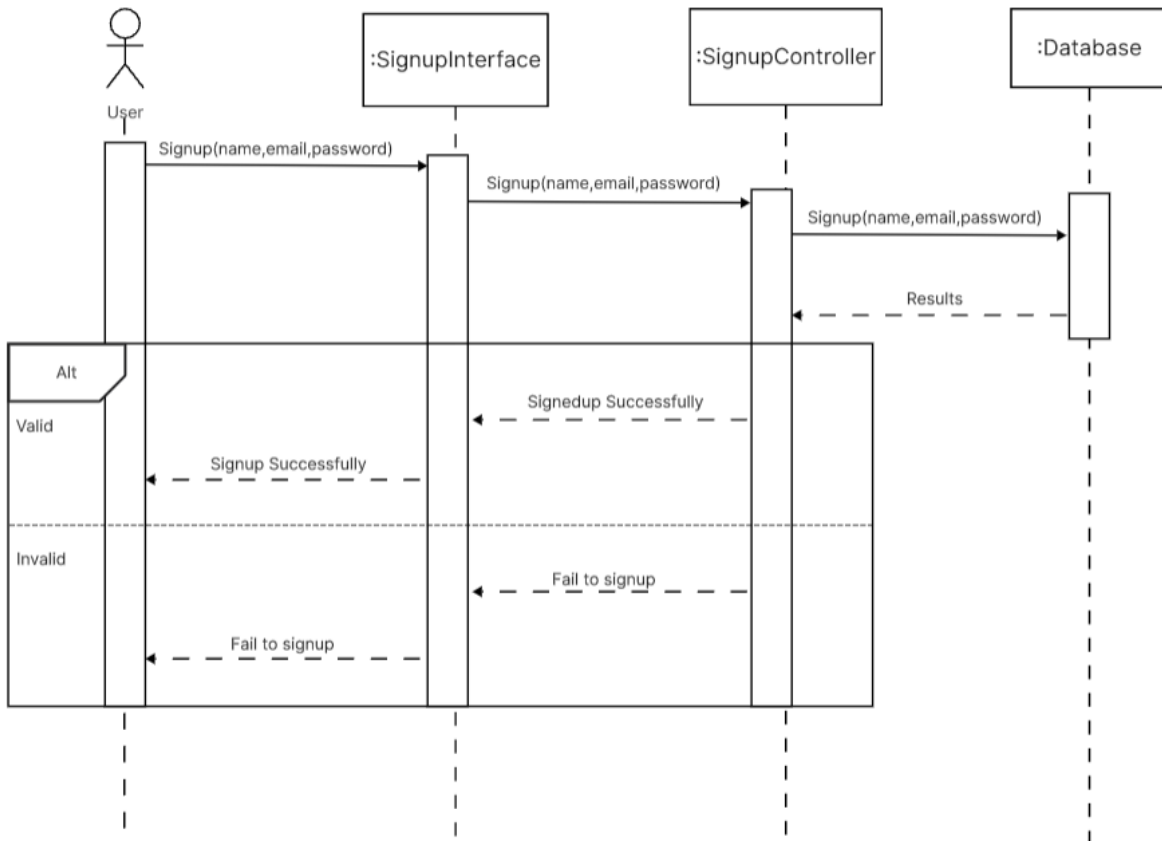
Software Design Description SDD is a representation of a software design used to communicate system design information to all stakeholders. It shows how the software system will be structured to satisfy the requirements.

3.2 Architectural Diagram

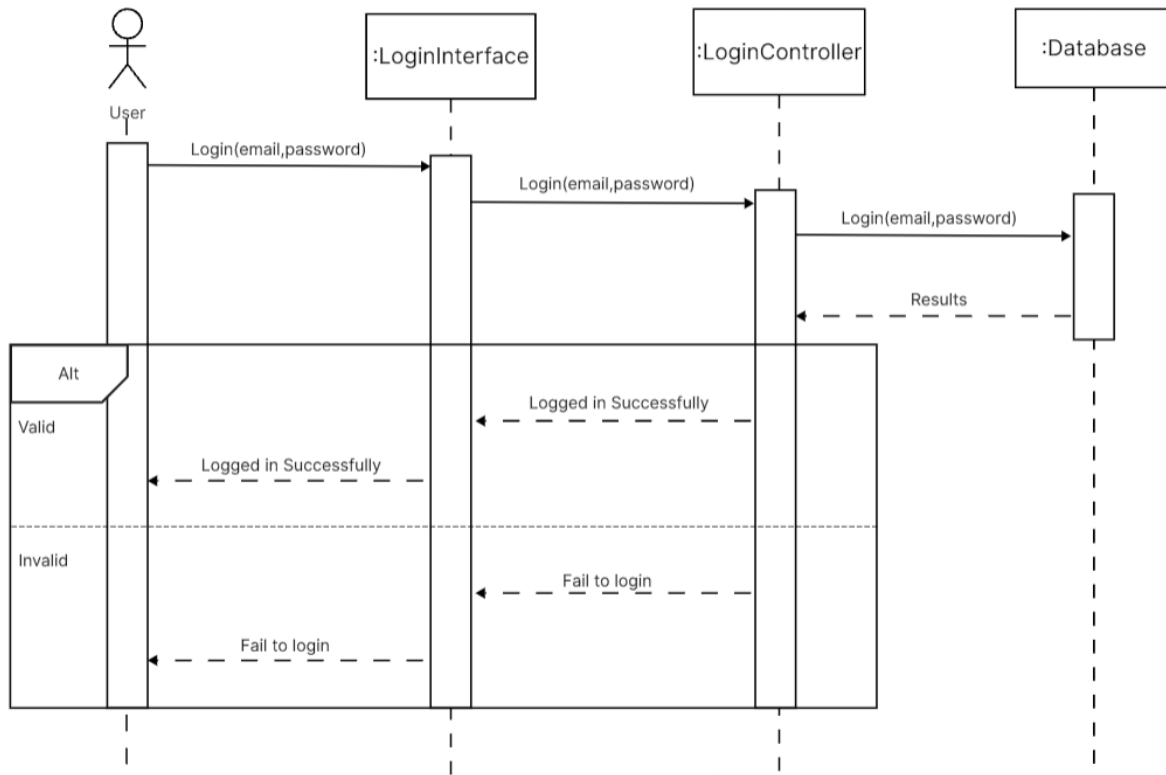


3.3 Sequence Diagrams:

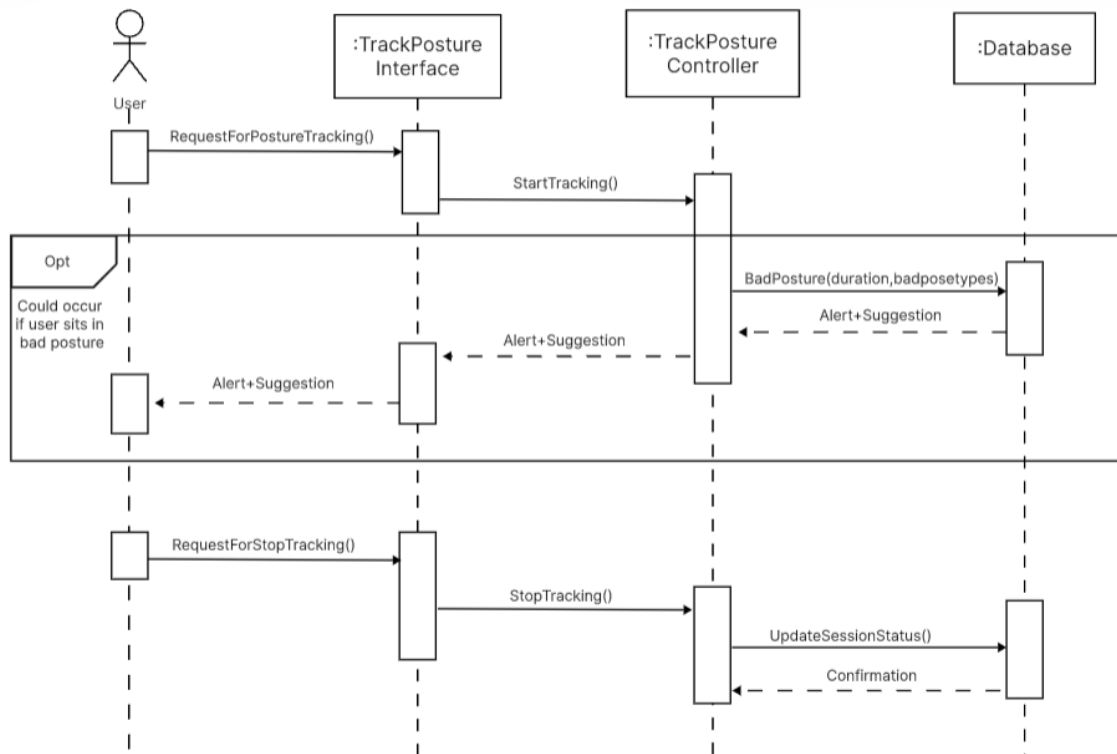
3.3.1 SignUp



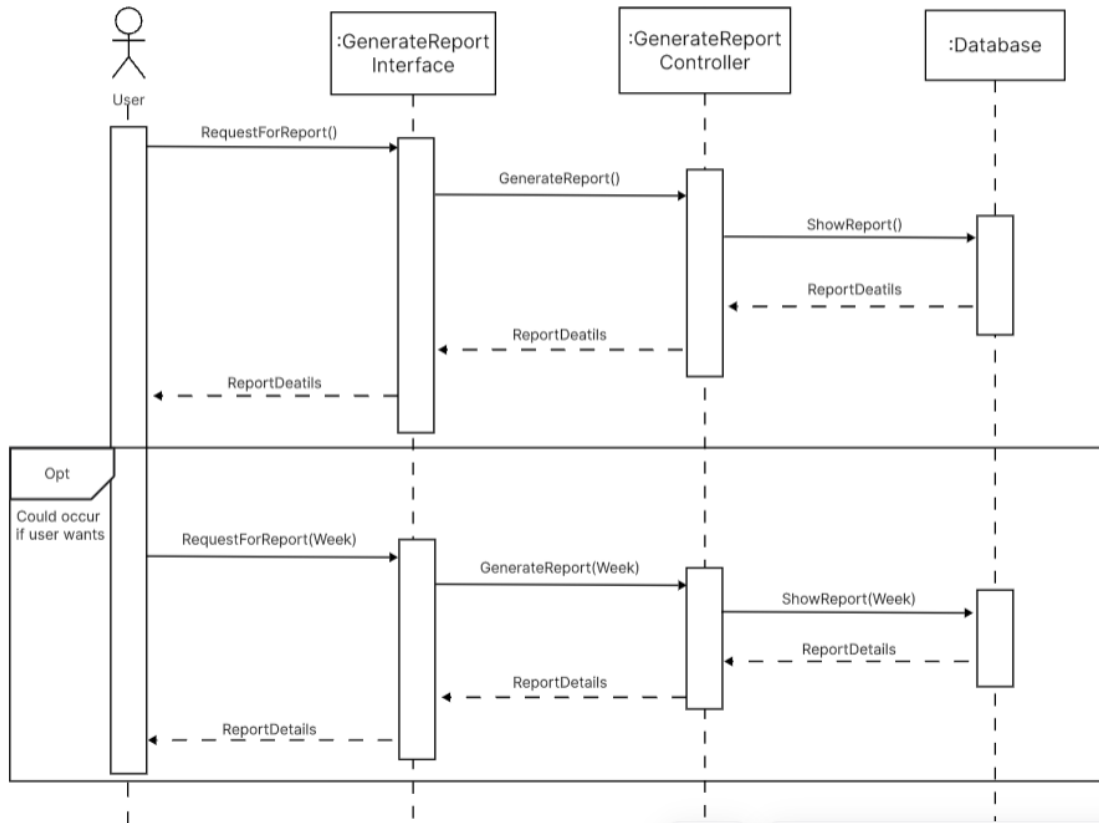
3.3.2 Login



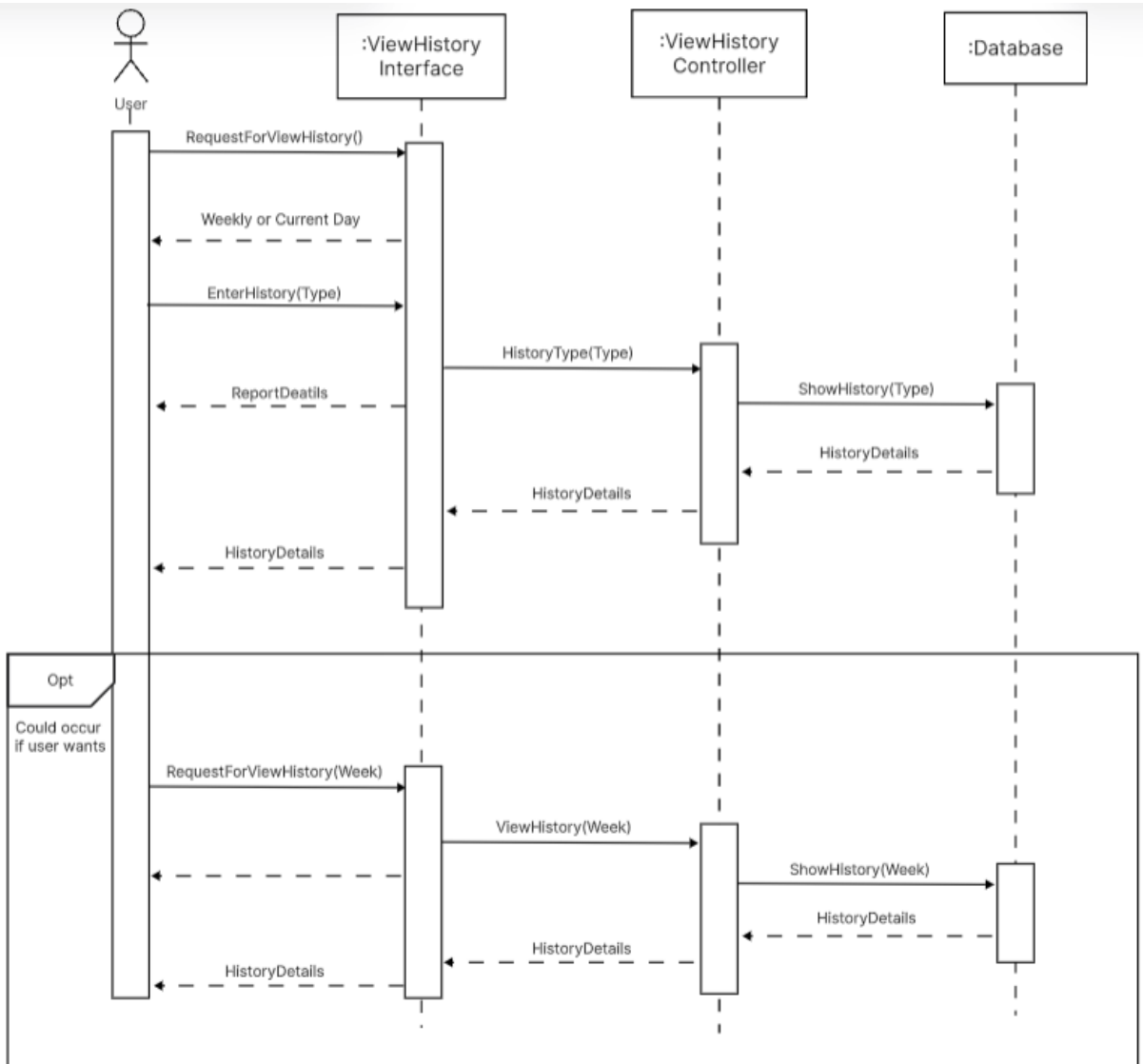
3.3.3 Track Posture



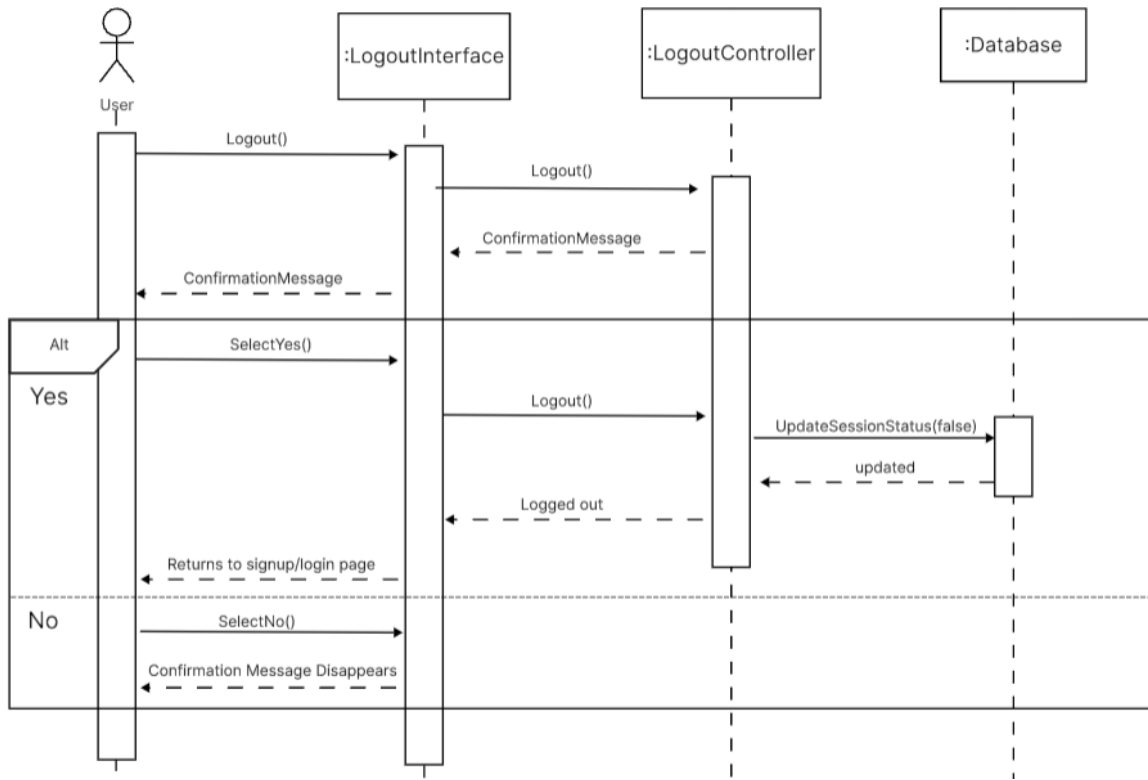
3.3.4 Generate Posture Report



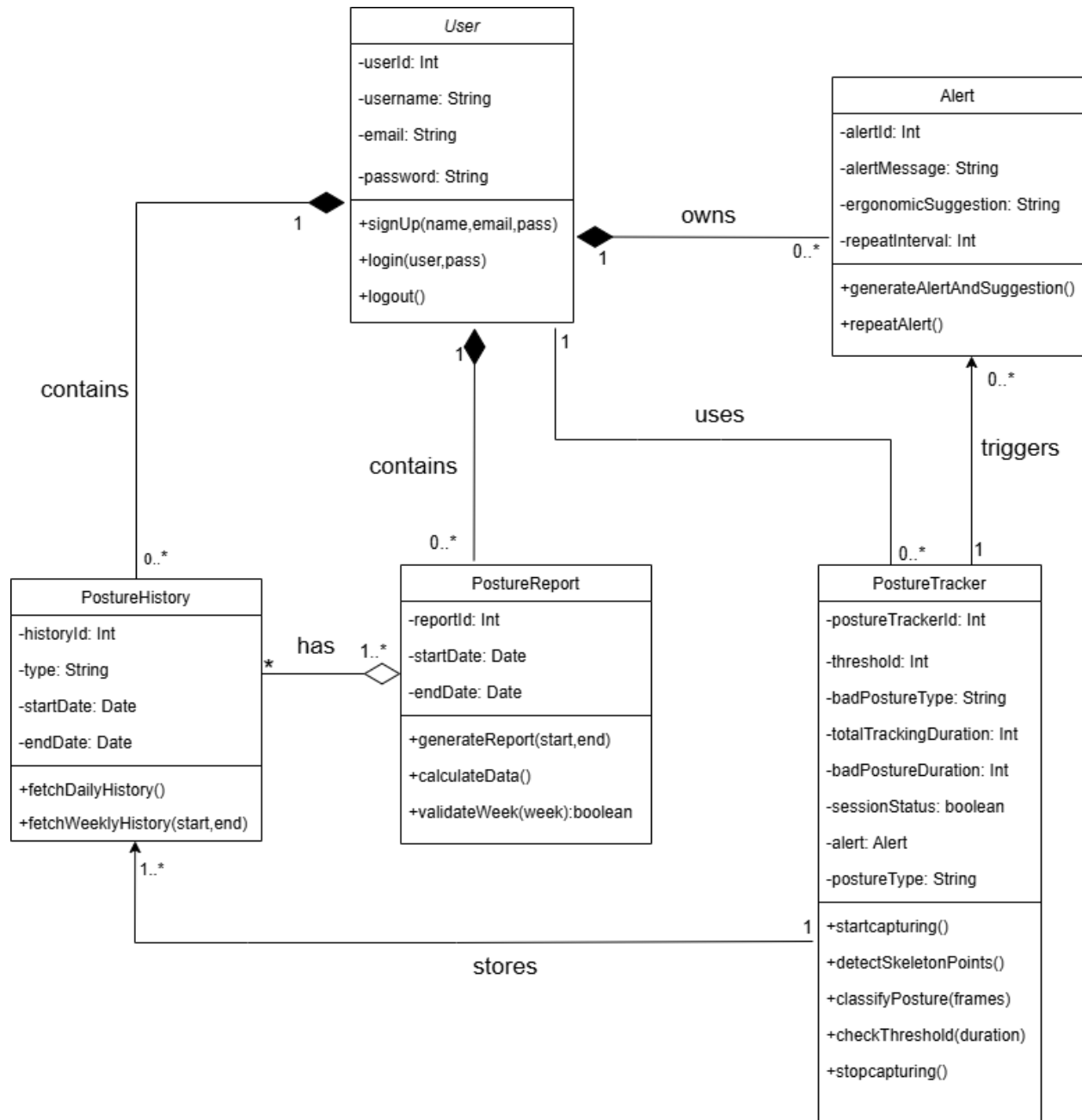
3.3.5 View Posture History



3.3.6 Logout



3.4 Class Diagram:



Interface Designs:

Signup

Login

Signup

Username :

Email:

Password:

create an account

Login

Username :

Password:

Login to Account

Dashboard

Start Tracking

Report

History

Logout

Home

Real time line graph

Stop Tracking

Report:

Home



Date:



Select Month

From: dd-mm-yyyy
To: dd-mm-yyyy

Total Monitoring Duration: -----
Total BAD Posture Duration: -----
BAD Posture %:-----
GOOD Posture %: -----
Most Frequent Bad Posture: -----

Generate

[Home](#)

History

[Current Day](#)[Weekly](#)[Home](#)

Date:



Select Month

History Table:

From: dd-mm-yyyy
To: dd-mm-yyyy

Date	Monitoring Duration	Posture Types	Bad Posture Duration	Bad Posture %	Good Posture %
dd-mm-yyyy	1h 52min	Leaning, forward head	50min	48%	52%
dd-mm-yyyy	2h	slouching	1h	50%	50%

[View](#)

CHAPTER NUMBER 4

SOFTWARE TESTING

4.1 Introduction

Software test document is a type of document under which test will determine whether system under test satisfies requirements or work correctly. Developing test cases can also help find problems in the requirement or design of an application.

4.2 Test Cases

4.2.1 Signup

ID: TC1

Name: Signup

Description: This testcase verifies that user creates an account successfully

Setup:

1. Network is stable
2. Website is opened

Instruction:

1. Click on "Signup"
2. Enter Email hubabmasood47@gmail.com
3. Enter password "Testing"
4. Click on "Creates an account"

Expected Result: main dashboard appears

Actual Result: Dashboard appears

Verdict : Pass

4.2.2 Login

ID: TC2

Name: Login

Description: This testcase verifies that user is logged in into his account successfully

Setup:

1. Network is stable
2. Account already exists

Instructions:

1. Click on "Login"
2. Enter Email as testing123@gmail.com
3. Enter password "hubtest"
4. Click on "Login" button

Expected Result: main dashboard appears

Actual result: Dashboard appears

Verdict: pass

4.2.3 Track Posture

ID: TC3

Name: Track posture

Description: This testcase checks that system detect the bad posture and generate alert and suggestion after threshold time of 1 min

Setup:

1. User sits Infront of laptop
2. Webcam is on and correctly working
3. User is logged in into the system

Instructions:

1. Click on "Track posture"
2. 3Sit in bad posture for one min
3. **Expected Result:** System shows alert and give suggestion to correct his posture
4. **Actual result:** alert and suggestion is being displayed

Verdict: pass

4.2.4 Generate Posture Report

ID: TC4

Name: Generate Posture Report

Description: This test case verifies that the system generates a weekly posture report successfully

Setup:

1. Network is stable
2. User is logged in into the system
3. User has used the application for at least one week

Instructions:

1. Click on "Generate Report"
2. Select a week from the available options Click on "Generate"

Expected Result: Weekly posture report is displayed showing total monitoring duration, total bad posture duration, bad posture percentage, and most frequent bad posture Actual

Result: Weekly report is displayed successfully

Verdict: Pass

4.2.5 View Posture History

ID: TC5

Name: View History

Description: This test case verifies that the system displays posture history correctly for current day and weekly view

Setup:

1. Network is stable
 2. User is logged in into the system
 3. User has at least one recorded posture session(for current Day)
- or
4. User has at least on week recorded Posture session

Instructions:

1. Click on "View History"
2. System asks for current day or weekly history Select "Current Day"

Expected Result: Current day posture history is displayed showing good posture duration and bad posture duration

Result: History is displayed successfully

Verdict: Pass

4.2.6 Logout

ID: TC6

Name: Logout

Description: This test case verifies that the user is logged out successfully after confirmation

Setup:

1. Network is stable
2. User is logged in into the system

Instructions:

1. Click on "Logout"
2. System shows confirmation dialog Click on "Yes"

Expected Result: User is logged out and login/signup page appears Actual

Result: User logged out and login/signup page is displayed

Verdict: Pass

Reference:

1. https://dspmuranchi.ac.in/pdf/Blog/srs_template-ieee.pdf